DSA03-Natural Language Processing

List of Lab Program

23. Develop a python program that evaluates the coherence of a given text.

Code:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import nltk
from nltk.tokenize import sent_tokenize
nltk.download('punkt_tab', quiet=True)
text = "The weather is sunny today. The sun is shining brightly. People enjoy the warm weather."
sentences = sent_tokenize(text)
print("Text:", text)
print("\nSentence Similarity Scores:")
if len(sentences) < 2:
    print("Not enough sentences to evaluate coherence.")
else:
    vectorizer = TfidfVectorizer()
    matrix = vectorizer.fit_transform(sentences)
    scores = []
    for i in range(len(sentences) - 1):
        score = cosine_similarity(
            matrix[i], matrix[i + 1]
        )[0][0]
```

```
        scores.append(score)
        print(f"Sentence {i+1} and {i+2}: {score:.2f}")
    average = sum(scores) / len(scores)
    print(f"\nAverage Coherence Score: {average:.2f}")
```

Output:

```
Text: The weather is sunny today. The sun is shining brightly. People enjoy the warm weather.

Sentence Similarity Scores:
Sentence 1 and 2: 0.25
Sentence 2 and 3: 0.09

Average Coherence Score: 0.17
```